PROBABILITY III

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Contents

Chapter 1

REFINING THE NOTION OF PROBABILITY

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Consider the random walk on the line; a drunk ant sits at the origin and takes steps of length 1 to the left or right with equal probability. We wish to obtain the probability that after an even n steps, the ant returns to the origin. Of course, the natural approach is to frame this problem in a discrete probability space.

Definition 1.1. A *discrete probability space* is a pair (Ω, q) where Ω is a non-empty finite or countably infinite set called the sample space, and $q : \Omega \to [0, 1]$ is a function satisfying $\sum_{\omega\in\Omega} q(\omega) = 1$. Any subset of Ω is termed an event, and the probability of an event $A \subseteq \Omega$ is defined as $\mathbb{P}(A) = \sum_{\omega\in A} q(\omega)$.

For the problem above, we can take the sample space to be

$$\Omega = \{\mathbf{x} = (x_0 = 0, x_1, x_2, \ldots, x_n) \in \mathbb{Z}^{n+1} \mid x_{i+1} - x_i \in \{-1, 1\} \text{ for all } 0 \le i < n\}, \tag{1.1}$$

and the mapping q to be $q(\mathbf{x}) = 2^{-n}$ for all $\mathbf{x} \in \Omega$. Then, the event that the ant returns to the origin after n steps is given by $A = \{\mathbf{x} \in \Omega \mid x_n = 0\}$, and we have

$$\mathbb{P}(A) = \sum_{\mathbf{x}\in A} q(\mathbf{x}) = 2^{-n}\,|A| = 2^{-n}\binom{n}{n/2}. \tag{1.2}$$

Reframing the problem, let us consider a random walk on the line starting at the origin, and continuing indefinitely. In this case, the sample space is given by

$$\Omega = \{\mathbf{x} = (x_0 = 0, x_1, x_2, \ldots) \in \mathbb{Z}^{\mathbb{N}} \mid x_{i+1} - x_i \in \{-1, 1\} \text{ for all } i \in \mathbb{N}\}. \tag{1.3}$$

This Ω is uncountably infinite. In such a case, we cannot define a probability measure by assigning a positive probability to each element of Ω. As another example of difficulty in dealing with uncountable sample spaces, consider the problem of choosing a real number uniformly at random from the interval $[0, 1]$. If we attempt to assign a positive probability to each element of $[0, 1]$, we would have to assign a probability of 0 to each element, since there are uncountably many elements in $[0, 1]$. However, this would lead to a total probability of 0, which is not valid. Of course, we know that the probability of choosing a specific element or a countable set of elements from $[0, 1]$ is 0, but there then arises the question of how to assign a probability to an uncountable set of elements, such as the Cantor set. We can work in a slightly different viewpoint.

Definition 1.2. A probability space is a tuple $(\Omega, \mathcal{F}, \mathbb{P})$ where Ω is a non-empty set called the sample space, $\mathcal{F}$ is the collection of all subsets of Ω $(\sim 2^{\Omega})$, and $\mathbb{P} : \mathcal{F} \to [0, 1]$ is a function satisfying $\mathbb{P}(\Omega) = 1$ and countable additivity, that is, for any countable collection of pairwise disjoint sets $A_1, A_2, \ldots \in \mathcal{F}$, we have

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i). \tag{1.4}$$

Using this definition, we look to define a $\mathbb{P}$ on subsets of $[0,1]$ that satisfies the above properties, along with the property that $\mathbb{P}([a,b]) = b-a$. Thus, we have $\mathbb{P}([0,1]) = 1$. If we have two disjoint intervals $I_1, I_2 \subseteq [0,1]$, then $\mathbb{P}(I_1 \cup I_2) = \mathbb{P}(I_1) + \mathbb{P}(I_2)$. This is a natural property to expect from a probability measure. What about a general set $S \subseteq [0,1]$? One can observe that if we have a sequence of intervals $I_1, I_2, \ldots$ such that $S \subseteq \bigcup_{k=1}^{\infty} I_k$, then we would have $\mathbb{P}(S) \le \sum_{k=1}^{\infty} \mathbb{P}(I_k)$. This is because the probability of S should be less than or equal to the probability of any set that contains S. Thus, we can define

$$\mathbb{P}_*(S) = \inf\left\{\sum_{k=1}^{\infty} |I_k| \;\middle|\; S \subseteq \bigcup_{k=1}^{\infty} I_k,\ I_k \subseteq [0,1]\right\}. \tag{1.5}$$

We have not verified that this is a valid definition of a probability measure, hence the subscript $*$. Via this, let us try computing $\mathbb{P}_*(\mathbb{Q} \cap [0,1])$. Enumerate the rationals in $[0,1]$ as $\{q_1, q_2, \ldots\}$. For any $\epsilon > 0$, we can choose intervals $I_k = [q_k - \epsilon/2^{k+1}, q_k + \epsilon/2^{k+1}] \cap [0,1]$ for each $k \in \mathbb{N}$. Note that $\bigcup_{k=1}^{\infty} I_k$ contains all the rationals in $[0,1]$, and $\sum_{k=1} |I_k| \le \sum_{k=1}^{\infty} \epsilon/2^k = \epsilon$. Since ϵ was arbitrary, we have $\mathbb{P}_*(\mathbb{Q} \cap [0,1]) = 0$. This is consistent with our intuition that the probability of choosing a rational number from $[0,1]$ is 0. So, $\mathbb{P}_*$ seems to be a reasonable definition of a probability measure on $([0,1], 2^{[0,1]})$. However, we will show later that there exists a set $\mathcal{A} \subseteq [0,1]$ such that $\mathbb{P}_*(\mathcal{A}) = \mathbb{P}_*([0,1] \setminus \mathcal{A}) = 1$, contradicting the property that $\mathbb{P}_*([0,1]) = 1$. We state the following theorem without proof.

Theorem 1.3. *There is no probability measure $\mathbb{P}$ on $([0,1], 2^{[0,1]})$ such that $\mathbb{P}([a,b]) = b-a$ for all $[a,b] \subseteq [0,1]$.*

If we remove the requirement that the probability measure of a subinterval $[a,b]$ is equal to its length, then we can define a probability measure on $([0,1], 2^{[0,1]})$. For example, we can define $\mathbb{P}(S) = 1$ if $1/2 \in S$ and $\mathbb{P}(S) = 0$ otherwise. This is a valid probability measure, but it is not very useful. Keeping the requirement that the probability measure of a subinterval $[a,b]$ is equal to its length, we can instead restrict the collection of subsets of $[0,1]$ on which we define the probability measure.

Definition 1.4. A *probability space* is a tuple $(\Omega, \mathcal{F}, \mathbb{P})$ where

- Ω is a non-empty set called the sample space,
- $\mathcal{F}$ is a collection of subsets of Ω satisfying the following properties:
  - $\Omega \in \mathcal{F}$,
  - if $A \in \mathcal{F}$, then $\Omega \setminus A \in \mathcal{F}$,
  - if $A_1, A_2, \ldots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$,
- $\mathbb{P} : \mathcal{F} \to [0,1]$ is a function satisfying:
  - $\mathbb{P}(\Omega) = 1$,
  - if $A_1, A_2, \ldots \in \mathcal{F}$ are pairwise disjoint, then $\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$.

Such an $\mathcal{F}$ is termd a *σ-algebra* on Ω, and such a $\mathbb{P}$ is termed a *probability measure* on $(\Omega, \mathcal{F})$.

1.1 σ-algebras

Example 1.5. Given Ω, trivial possible choices for $\mathcal{F}$ are $\{\emptyset, \Omega\}$ and 2^{Ω}. A more interesting example is the following: suppose Ω is uncountable. Then $\mathcal{F} = \{A \subseteq \Omega \mid A \text{ is countable or } \Omega \setminus A \text{ is countable}\}$ is a σ-algebra on Ω.

Let $S \subseteq 2^\Omega$ be a collection of subsets of Ω. We claim that there exists a notion of the smallest σ-algebra containing S. This is called the σ-algebra generated by S, and is denoted by $\sigma(S)$.

Lemma 1.6. *For any collection $S \subseteq 2^\Omega$, there exists a unique smallest σ-algebra containing S satisfying*

$$\sigma(S) = \bigcap \{\mathcal{F} \subseteq 2^\Omega \mid \mathcal{F} \text{ is a } \sigma\text{-algebra and } S \subseteq \mathcal{F}\}. \tag{1.6}$$

Proof. It is easy to check that $\sigma(S)$ is indeed a σ-algebra containing S. If $\mathcal{F}$ is any σ-algebra containing S, then $\sigma(S) \subseteq \mathcal{F}$. Thus, $\sigma(S)$ is the unique smallest σ-algebra containing S. ■

Definition 1.7. Let (X, d) be a metric space (or a topological space), and let $\mathcal{S} \subseteq 2^X$ be the colletion of all open sets in X. Then, the σ-algebra $\sigma(\mathcal{S})$ is called the *Borel σ-algebra* on X, and its elements are called *Borel sets*.

We will show that there exists a probability measure on $([0,1], \mathcal{B})$, where $\mathcal{B}$ is the Borel σ-algebra on $[0,1]$, which extends the notion of length of intervals to Borel sets.

Let us return to the problem of indefinite random walks on the line. Let $\Omega = \{\omega = (\omega_1, \omega_2, \ldots) \mid \omega_i \in \{0,1\}\}$, where $\omega_i = 0$ corresponds to a step to the left and $\omega_i = 1$ corresponds to a step to the right. If we let $A = \{0,1\}$, then $\Omega = A^{\mathbb{N}}$; consider the discrete topology on A, and the product topology on Ω. If we let

$$\mathcal{S} = \{U_1 \times U_2 \times \cdots \times U_k \times A \times A \cdots \mid U_i = \{0\} \text{ or } \{1\}\} \tag{1.7}$$

then one can show that the σ-algebra generated by $\mathcal{S}$ is the Borel σ-algebra on Ω. We can set $\mathcal{F} = \sigma(\mathcal{S})$, and define a probability measure $\mathbb{P}$ on $(\Omega, \mathcal{F})$ by setting $\mathbb{P}(U_1 \times U_2 \times \cdots \times U_k \times A \times A \cdots) = 2^{-k}$ for all $U_i = \{0\}$ or $\{1\}$.

July 24th.

Example 1.8. Let (X, d) be a separable metric space (that is, there exists a countable dense subset). The Borel σ-algebra on X is then

$$\mathcal{B}(X) = \sigma(\{B(p,r)\}_{p \in X, r > 0}). \tag{1.8}$$

Note that the usual containment is $\sigma(\{B(p,r)\}_{p \in X, r>0}) \subseteq \mathcal{B}(X)$ for a metric space X. So, it suffices to show that if $V \in \tau$, the topology on X, then $V \in \sigma(\{B(p,r)\}_{p \in X, r>0})$. Let Q be a countable dense subset of X. Then, for any $V \in \tau$, we have

$$V = \bigcup_{n=1}^{\infty} B(q_n, r_n), \tag{1.9}$$

where $q_n \in Q \cap V$ and $r_n > 0$ (where r_n is the largest possible radius such that $B(q_n, r_n) \subseteq V$). Since $\sigma(\{B(p,r)\}_{p \in X, r>0})$ is a σ-algebra, we have $V \in \sigma(\{B(p,r)\}_{p \in X, r>0})$. Thus, we have $\mathcal{B}(X) = \sigma(\{B(p,r)\}_{p \in X, r>0})$.

Example 1.9. Let $(\Sigma, \mathcal{F}, \mathbb{P})$ be a probability space.

- $\mathcal{F}$ is closed under countable unions, countable intersections, differences, and symmetric differences. Countable unions and intersections follow from the definition of a σ-algebra. If $A, B \in \mathcal{F}$, then $A \setminus B = A \cap (\Omega \setminus B) \in \mathcal{F}$. Finally, $A \triangle B = (A \setminus B) \cup (B \setminus A) \in \mathcal{F}$.
- Let $(\Sigma, \mathcal{F}, \mathbb{P})$ be a probability space, and let $A_1, A_2, \ldots \in \mathcal{F}$. Define

$$\limsup_{n \to \infty} A_n := \{x \in \Omega \mid x \text{ belongs to infinitely many } A_n\} \tag{1.10}$$

and

$$\liminf_{n \to \infty} A_n := \{x \in \Omega \mid x \text{ belongs to all but finitely many } A_n\}. \tag{1.11}$$

Then, both the lim sup and lim inf belong in $\mathcal{F}$. To see this, simply note that

$$\limsup_{n\to\infty} A_n = \bigcap_{n=1}^{\infty}\bigcup_{k=n}^{\infty} A_k, \quad \liminf_{n\to\infty} A_n = \bigcup_{n=1}^{\infty}\bigcap_{k=n}^{\infty} A_k. \tag{1.12}$$

- One may show that $\mathbb{P}(\emptyset) = 0$, $\mathbb{P}(A) \le \mathbb{P}(B)$ if $A \subseteq B$, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$, and $\mathbb{P}(\bigcup A_n) \le \sum \mathbb{P}(A_n)$ for any countable collection of sets $A_1, A_2, \ldots \in \mathcal{F}$.
- For a nested sequence of sets $A_1 \subseteq A_2 \subseteq \cdots$ in $\mathcal{F}$, let $A = \bigcup_{n=1}^{\infty} A_n$ (this is usually denoted by $A_n \uparrow A$). Then, $\mathbb{P}(A) = \lim_{n\to\infty} \mathbb{P}(A_n)$. To this end, let $B_n = A_n \setminus A_{n-1}$ for $n > 1$, and $B_1 = A_1$. Then $A = \bigcup_{n=1}^{\infty} B_n$. Thus,

$$\mathbb{P}(A) = \sum_{n=1}^{\infty} \mathbb{P}(B_n) = \lim_{N\to\infty}\sum_{n=1}^{N} \mathbb{P}(B_n) = \lim_{N\to\infty}\sum_{n=2}^{N} \mathbb{P}(A_n) - \mathbb{P}(A_{n-1}) + \mathbb{P}(A_1) = \lim_{N\to\infty} \mathbb{P}(A_N). \tag{1.13}$$

1.1.1 Other Systems of Sets

The following are other interesting collection of subsets of Ω besides the Borel σ-algebra.

1. *π-system*: a collection of subsets $S \subseteq \Omega$ such that $A, B \in S$ implies $A \cap B \in S$.
2. *λ-system*: a collection Λ of subsets of Ω such that $\Omega \in \Lambda$, if $A, B \in \Lambda$ with $A \subseteq B$, then $B \setminus A \in \Lambda$, and if $A_1, A_2, \ldots \in \Lambda$, then $\bigcup_{n=1}^{\infty} A_n \in \Lambda$.
3. *algebra*: a collection $\mathcal{A}$ of subsets of Ω such that $\Omega \in \mathcal{A}$, if $A \in \mathcal{A}$, then $\Omega \setminus A \in \mathcal{A}$, and if $A, B \in \mathcal{A}$, then $A \cup B \in \mathcal{A}$.

We had defined $\omega(S)$, the σ-algebra generated by a collection of subsets $S \subseteq 2^{\Omega}$. We can similarly define $\pi(S)$, the π-system generated by S, $\lambda(S)$, the λ-system generated by S, and $\mathcal{A}(S)$, the algebra generated by S.

Theorem 1.10 (*Sierpiński-Dynkin π-λ Theorem*)**.** *Let $S, \mathcal{F} \subseteq 2^{\Omega}$ be collections of subsets of Ω. Then,*

1. *$\mathcal{F}$ is a σ-algebra if and only if $\mathcal{F}$ is both a π-system and a λ-system.*
2. *If S is a π-system, then $\lambda(S) = \sigma(S)$.*

Proof.

1. If $\mathcal{F}$ is a σ-algebra, then it is trivially both a π-system and a λ-system. Conversely, if $\mathcal{F}$ is both a π-system and a λ-system, we need to verify the axioms of a σ-algebra. We have $\Omega \in \mathcal{F}$ since $\mathcal{F}$ is a λ-system. If $A \in \mathcal{F}$, then $\Omega \setminus A \in \mathcal{F}$ since $\mathcal{F}$ is a λ-system. Finally, if $A_1, A_2, \ldots \in \mathcal{F}$, then we can write $B_n = \bigcup_{i=1}^{n} A_i \in \mathcal{F}$ since $A_i^c \in \mathcal{F}$ and $B_n^c = \bigcap_{i=1}^{n} A_i^c \in \mathcal{F}$ since $\mathcal{F}$ is a π-system. Then, $B_n \uparrow B = \bigcup_{i=1}^{\infty} A_i$, and since $\mathcal{F}$ is a λ-system, we have $B \in \mathcal{F}$. Thus, $\mathcal{F}$ is a σ-algebra.
2. Let S be a π-system. Since $\lambda(S) \subseteq \sigma(S)$, it suffices to show that $\lambda(S)$ is a π-system. Fix $A \in S$, and define $\mathcal{C}_A = \{B \in \lambda(S) \mid A \cap B \in \lambda(S)\}$. Note that $S \subseteq \mathcal{C}_A$ because S is a π-system. We show that $\mathcal{C}_A$ is a λ-system. We have $\Omega \in \mathcal{C}_A$ since $A \cap \Omega = A \in \lambda(S)$. If $B_1, B_2 \in \mathcal{C}_A$ with $B_1 \subseteq B_2$, then $A \cap B_1 \subseteq A \cap B_2$, and since $A \cap B_2 \in \lambda(S)$, we have $(A \cap B_2) \setminus (A \cap B_1) = A \cap (B_2 \setminus B_1) \in \lambda(S)$, and thus $B_2 \setminus B_1 \in \mathcal{C}_A$. Finally, if $B_1, B_2, \ldots$ are in $\mathcal{C}_A$, then $A \cap B_n \in \lambda(S)$ for all n, and thus $\bigcup (A \cap B_n) = A \cap (\bigcup B_n) \in \lambda(S)$, and hence $\bigcup B_n \in \mathcal{C}_A$. Thus, $\mathcal{C}_A$ is a λ-system. Since $\lambda(S)$ is the smallest λ-system containing S, we have $\lambda(S) = \mathcal{C}_A$, and hence for any $B, C \in S$, we have $B, C \in S$ implies $B \cap C = A \cap C \in S$. Thus, $\lambda(S)$ is a π-system.

■

Lemma 1.11. *Let S be a π-system, and let $\mathbb{P}$ and $\mathbb{Q}$ be two probability measures on $(\Sigma, \sigma(S))$ such that $\mathbb{P}(A) = \mathbb{Q}(A)$ for all $A \in S$. Then, $\mathbb{P}(A) = \mathbb{Q}(A)$ for all $A \in \sigma(S)$.*

Proof. Let $\mathcal{C} = \{B \in \sigma(S) \mid \mathbb{P}(B) = \mathbb{Q}(B)\}$. Then, $S \subseteq \mathcal{C}$, and we show that $\mathcal{C}$ is a λ-system. We have $\Omega \in \mathcal{C}$ since $\mathbb{P}(\Omega) = 1 = \mathbb{Q}(\Omega)$. If $B_1, B_2 \in \mathcal{C}$ with $B_1 \subseteq B_2$, then $\mathbb{P}(B_2 \setminus B_1) = \mathbb{P}(B_2) - \mathbb{P}(B_1) = \mathbb{Q}(B_2) - \mathbb{Q}(B_1) = \mathbb{Q}(B_2 \setminus B_1)$, and hence $B_2 \setminus B_1 \in \mathcal{C}$. Finally, if $B_1, B_2, \ldots$ are in $\mathcal{C}$, then $\mathbb{P}(A) = \lim_{n\to\infty} \mathbb{P}(A_n)$, and since $\mathbb{Q}(A) = \lim_{n\to\infty} \mathbb{Q}(A_n)$, we have $\mathbb{P}(A) = \mathbb{Q}(A)$. ■

July 28th.

Example 1.12. We wish to obtain an example of S which is *not* a π-system, and two probability measures $\mathbb{P}$ and $\mathbb{Q}$ on $(\Sigma, \sigma(S))$ such that $\mathbb{P}(A) = \mathbb{Q}(A)$ for all $A \in S$, but $\mathbb{P}(A) \neq \mathbb{Q}(A)$ for some $A \in \sigma(S)$. Let $\Omega = \{1, 2, 3, 4\}$, and let $S = \{\{1, 2\}, \{2, 3\}\}$. Then, $\sigma(S) = 2^{\Omega}$. Define $\mathbb{P}$ and $\mathbb{Q}$ on $(\Omega, \sigma(S))$ by

$$\mathbb{P}(\{1\}) = \mathbb{P}(\{2\}) = \mathbb{P}(\{3\}) = \mathbb{P}(\{4\}) = 1/4, \tag{1.14}$$

and

$$\mathbb{Q}(\{1\}) = \mathbb{Q}(\{3\}) = 1/6, \quad \mathbb{Q}(\{2\}) = \mathbb{Q}(\{4\}) = 1/3. \tag{1.15}$$

Then, $\mathbb{P}(\{1, 2\}) = \mathbb{Q}(\{1, 2\}) = 1/2$ and $\mathbb{P}(\{2, 3\}) = \mathbb{Q}(\{2, 3\}) = 1/2$, but $\mathbb{P}(\{3\}) = 1/4 \neq 1/6 = \mathbb{Q}(\{3\})$.

Example 1.13. Let $\Omega \neq \emptyset$, and A_λ be a collection of subsets of Ω indexed by $\lambda \in \Lambda$. Consider the σ-algebra $\sigma(\{A_\lambda\}_{\lambda\in\Lambda})$, and pick $B \in \sigma(\{A_\lambda\}_{\lambda\in\Lambda})$. We will show that there exists a countable subcollection $\{A_{\lambda_n}\}_{n=1}^{\infty}$ such that $B \in \sigma(\{A_{\lambda_n}\}_{n=1}^{\infty})$. Let $\mathcal{C} = \{B \in \sigma(\{A_\lambda\}_{\lambda\in\Lambda}) \mid B \in \sigma(\{A_{\lambda_n}\}_{n=1}^{\infty})$ for some countable subcollection $\{A_{\lambda_n}\}_{n=1}^{\infty}\}$. Then, $\mathcal{C}$ is a σ-algebra containing $\{A_\lambda\}_{\lambda\in\Lambda}$, and hence $\sigma(\{A_\lambda\}_{\lambda\in\Lambda}) \subseteq \mathcal{C}$. Thus, $B \in \mathcal{C}$, and hence there exists a countable subcollection $\{A_{\lambda_n}\}_{n=1}^{\infty}$ such that $B \in \sigma(\{A_{\lambda_n}\}_{n=1}^{\infty})$.

1.2 The Lebesgue Measure

Theorem 1.14. *There exists a unique Borel measure λ on $[0, 1]$ such that $\lambda(I) = |I|$ for all intervals $I \subseteq [0, 1]$.*

Proof. For the π-system $S = \{(a, b] \mid 0 \leq a < b \leq 1\}$, we know that $\sigma(S)$ is a Boreal σ-algebra on $[0, 1]$. If λ and μ are two Borel measures on $[0, 1]$ such that $\lambda(I) = \mu(I) = |I|$ for all intervals $I \subseteq [0, 1]$, then by the previous lemma, we have $\lambda(A) = \mu(A)$ for all $A \in \sigma(S)$. Thus, the Borel measure is unique.

To show the existence of such a measure, we will define a pseudomeasure λ_* that satisfies the interval property, and then show that it is a measure on the Borel σ-algebra. For any $E \subseteq [0, 1]$, define

$$\lambda_*(E) = \inf\left\{\sum_{i=1}^{\infty} |I_i| \;\middle|\; I_i \text{ are intervals and } E \subseteq \bigcup_{i=1}^{\infty} I_i\right\} \tag{1.16}$$

Infer that $0 \leq \lambda_*(E) \leq 1$ for all $E \subseteq [0, 1]$. Moreover, let A and B be two sets in $[0, 1]$. Then, for any $\varepsilon > 0$, we have

$$\lambda_*(A \cup B) \leq \sum_{k=1}^{\infty} |I_k| + \sum_{k=1}^{\infty} |J_k| \leq \lambda_*(A) + \varepsilon + \lambda_*(B) + \varepsilon \implies \lambda_*(A \cup B) \leq \lambda_*(A) + \lambda_*(B). \tag{1.17}$$

Finally, let $(a, b] \subseteq [0, 1]$ be an interval. Then, $\lambda_*((a, b]) = b - a$. To see this, suppose $(a, b] \subseteq \bigcup_{k=1}^{\infty} I_k$ for some intervals I_k. Now consider $[a + \varepsilon, b]$ for some $\varepsilon > 0$ such that $[a + \varepsilon, b] \subseteq \bigcup_{k=1}^{\infty} I_k$. This is a compact subinterval, so we can extract a finite subcover $[a + \varepsilon, b] \subseteq \bigcup_{k=1}^{N} I_k$. Then,

$$b - a - \varepsilon = |[a + \varepsilon, b]| \leq \sum_{k=1}^{N} |I_k| \leq \sum_{k=1}^{\infty} |I_k| \implies b - a \leq \lambda_*((a, b]). \tag{1.18}$$

Now, since $(a,b] \subseteq (a, b+\varepsilon)$, a similar line of reasoning shows that $\lambda_*((a,b]) \leq b-a$. Thus, we have $\lambda_*((a,b]) = b-a$. The property of finite subadditivity also implies the property of countable subadditivity (show this). We now construct the σ-algebra $\mathcal{F}$ on which λ_* is a measure. We pick

$$\mathcal{F} = \{A \subseteq [0,1] \mid \lambda_*(E) = \lambda_*(E \cap A) + \lambda_*(E \cap A^c) \text{ for all } E \subseteq [0,1]\}. \tag{1.19}$$

We show that $\mathcal{F}$ is a σ-algebra containing the Borel σ-algebra, and that λ_* is a measure on $\mathcal{F}$. It is left as a minor exercise to show that $\mathcal{F}$ is closed under intersections and complements (making $\mathcal{F}$ a π-system). We show $\mathcal{F}$ is a λ-system. Let $A, B \in \mathcal{F}$. Then $B \setminus A = B \cap A^c$ is in $\mathcal{F}$. Then, for any $E \subseteq [0,1]$, we have

$$\begin{aligned} \lambda_*(E) &= \lambda_*(E \cap A_n) + \lambda_*(E \cap A_n^c) && (1.20)\\ &\geq \lambda_*(E \cap A_n) + \lambda_*(E \cap A^c), && (1.21) \end{aligned}$$

because $A^c \subseteq A_n^c$. Since $A_n \uparrow A$, we have $E \cap A_n \uparrow E \cap A$, and hence

$$\lambda_*(E \cap A_n) \uparrow \lambda_*(E \cap A).$$

Passing to the limit gives

$$\lambda_*(E) \geq \lambda_*(E \cap A) + \lambda_*(E \cap A^c).$$

On the other hand, subadditivity of the outer measure gives

$$\lambda_*(E) \leq \lambda_*(E \cap A) + \lambda_*(E \cap A^c).$$

Therefore,

$$\lambda_*(E) = \lambda_*(E \cap A) + \lambda_*(E \cap A^c),$$

so $A \in \mathcal{F}$. ■

Index